Geometric sequences

1 Explain how the common ratio of a geometric sequence can be found.

2 Find the common ratio of the following geometric sequences:

a $2, -16, 128, -1024, \ldots$

b $-9, 8.1, -7.29, 6.561, \ldots$

c $-70.4, -17.6, -4.4, -1.1, \ldots$

3 Write down the next two terms for the following sequences:

a $4, 12, 36, \ldots$

b $12, -48, 192, \ldots$

c $1, \dfrac{3}{4}, \dfrac{9}{16}, \ldots$

d $-6, 9, -\dfrac{27}{2}, \ldots$

4 Consider the first four terms in this geometric sequence: $-8, -16, -32, -64, \ldots$

a Evaluate:

i $\dfrac{T_2}{T_1}$

ii $\dfrac{T_3}{T_2}$

iii $\dfrac{T_4}{T_3}$

b Find the value of T_5.

5 For each of the following, write the first four terms in the geometric progression:

a The first term is 6 and the common ratio is 4.

b The first term is 7 and the common ratio is -2.

c The first term is -2 and the common ratio is 3.

d The first term is 1.3 and the common ratio is -4.

e The first term is $700\,000$ and the common ratio is 1.04.

6 Consider the sequence: $4, -28, 224, -1372, \ldots$
Is the sequence arithmetic, geometric or neither?

7 For each of the following:

i Determine if the sequence is arithmetic or geometric. Explain your answer.

ii Find the common ratio or common difference, whichever is applicable.

a $11, -99, 891, -8019, \ldots$

b $2, 6, 10, 14, \ldots$

8 Find the missing terms in the following geo[redacted]c progressions:

a $-5, x, -80, 320, y$

b $a, b, \frac{3}{25}, -\frac{3}{125}, c$

9 Suppose $t_1, t_2, t_3, t_4, t_5, \ldots$ is a geometric sequence.
Is $t_1, t_3, t_5, \ldots$ also a geometric sequence? Explain your answer.

10 If the first term of a sequence is 27 and the common ratio is $\frac{1}{3}$, find the 10th term.

11 If the first term of a sequence is 1.2 and the common ratio is -5, find the 4th term.

12 If the first term of a sequence is $400\,000$ and the common ratio is 1.12, find the 3rd term.

13 Find the common ratio of the geometric sequence whose first term is 5 and third term is 320.

14 In a geometric progression, $T_7 = \frac{64}{81}$ and $T_8 = \frac{128}{243}$.

a Find the value of r, the common ratio in the sequence.

b Find the first three terms of the geometric progression.

15 Find three values between 18 and $\frac{32}{9}$ such that the five terms form successive terms in a geometric progression.

16 Find three consecutive positive terms of a geometric progression if they have a product of 125 and the third term is 9 times the first term.

17 Consider the finite sequence: $4, 4\sqrt{2}, 8, \ldots, 256$

a Find the common ratio.

b Find T_6.

c Solve for n, the number of terms in the sequence.

Explicit rules

18 For each of the given explicit rules:

i List the first four terms of the sequence.

ii State the common ratio.

a $T_n = 3 \times 4^{n-1}$

b $T_n = -4 \times (-3)^{n-1}$

19 In a geometric progression, $T_4 = 32$ and $T_6 = 128$.

a Find the value of r, the common ratio in the sequence.

b For the case where r is postive, find the value of a, the first term in the sequence.

c Consider the sequence in which the first term is positive.
Find an expression for T_n, the nth term of this sequence.

20 The nth term of a geometric progression is given by the equation $T_n = 25 \times \left(\frac{1}{5}\right)^{n-1}$.

a Complete the table of values:

b Find the common ratio.

n	1	2	3	4	10
T_n					

21 The nth term of a geometric progression is given by the equation $T_n = -72 \times \left(-\frac{4}{3}\right)^{n-1}$.

a Complete the table of values:

b Find the common ratio.

n	1	2	3	4	6
T_n					

22 Each of the given tables of values represents terms in a geometric sequence:

i Find r, the common ratio between consecutive terms.

ii Write a simplified expression for the nth term of the sequence, T_n.

iii Find the missing term in the table.

a

n	1	2	3	4	10
T_n	5	40	320	2560	

b

n	1	2	3	4	12
T_n	7	-21	63	-189	

c

n	1	3	6	9	11
T_n	-5	-45	-1215	$-32\,805$	

d

n	1	2	3	4	7
T_n	-2	$-\frac{16}{3}$	$-\frac{128}{9}$	$-\frac{1024}{27}$	

23 The values in the table show the nth term in a geometric sequence for consecutive values of n. Complete the missing values in the table:

a

n	1	2	3	4	5
T_n	5			-320	

b

n	1	2	3	4	5
T_n	-27			-64	

24 The given table of values represents terms in a geometric sequence:

a Find r, the common ratio between consecutive terms.

b Write a simplified expression for the nth term of the sequence, T_n.

n	1	4	9
T_n	-9	576	$-589\,824$

25 Consider the following sequence: $2, -16, 12$[redacted]$024, \ldots$

a State the general expression for the nth term of the sequence.

b Find the next three terms of the sequence.

c Find T_{11}.

26 Consider the following sequence: $-0.3, -1.5, -7.5, -37.5, \ldots$

a Find the formula for the nth term of the sequence.

b Find the next three terms of the sequence.

Recursive rules

27 Consider the sequence: $9000, 1800, 360, 72, \ldots$
Write a recursive rule for T_n in terms of T_{n-1} and an initial condition for T_1.

28 The first term of a geometric sequence is 6. The fourth term is 384.

a Find the common ratio, r, of this sequence.

b State the recursive rule, T_{n+1}, that defines this sequence.

29 The first term of a geometric sequence is 5. The third term is 80.

a Find the possible values of the common ratio, r.

b Find the recursive rule, T_n, that defines the sequence with a positive common ratio.

c Find the recursive rule, T_n, that defines the sequence with a negative common ratio.

30 The third term of a geometric sequence is 16. The sixth term is 128.

a Find the common ratio, r, of this sequence.

b Find the first term of this sequence.

c State the recursive rule, T_{n+1}, that defines this sequence.

31 The third term of a geometric sequence is 2[redacted]he seventh term is 4.

- **a** Find the common ratio, r of this sequence.
- **b** Find the first term of this sequence.
- **c** Write a recursive rule, T_n, that defines the sequence with a positive common ratio.
- **d** Write a recursive rule, T_n, that defines the sequence with a negative common ratio.

32 Consider the sequence: $-1, -7, -49, \ldots$

- **a** Find the next term of the sequence.
- **b** Find the 5th term of the sequence.
- **c** Find the 6th term of the sequence.

Sequences on the coordinate plane

33 The nth term of a geometric progression is given by the equation $T_n = 2 \times 3^{n-1}$

- **a** Complete the table of values:
- **b** Find the common ratio between consecutive terms.

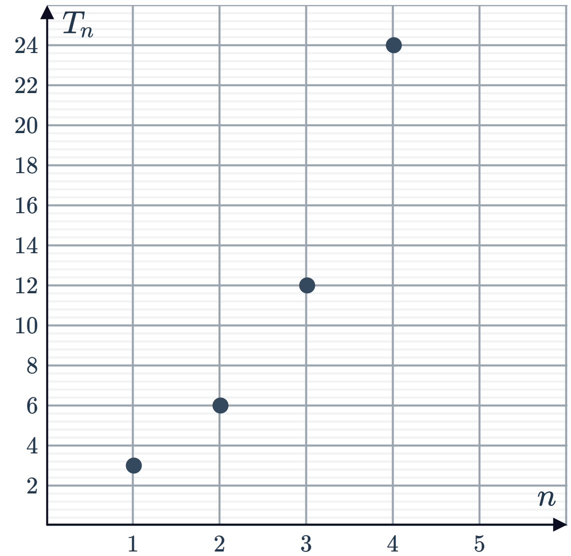
n	1	2	3	4	10
T_n					

- **c** Plot the first four points in the table on a coordinate plane.
- **d** State whether the joined points would form a straight line or a curve.

34 Consider the sequence: $2, 6, 18, 54, \ldots$

- **a** Plot the first four terms on a coordinate plane.
- **b** Is the sequence arithmetic or geometric?
- **c** Write a recursive rule for T_n in terms of T_{n-1} and an initial condition for T_1.

35 The plotted points represent terms in a geo[redacted]c sequence:



a Complete the table of values for the given points:

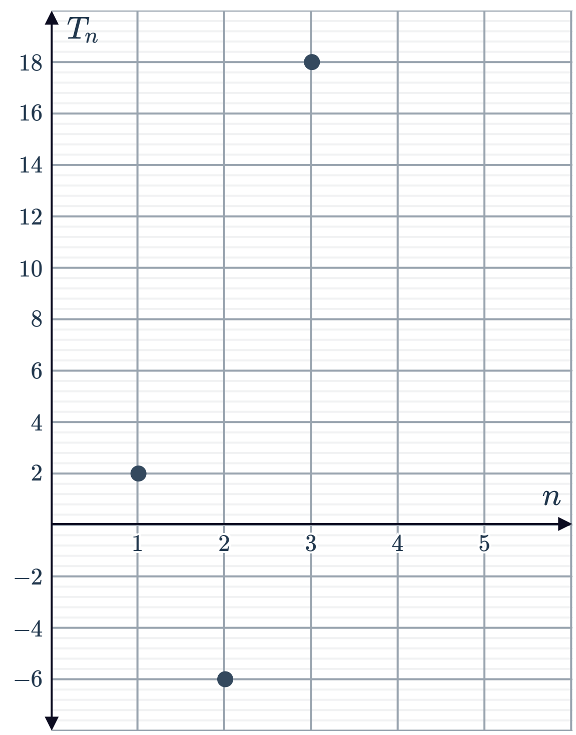
n	1	2	3	4
T_n				

b Find r, the common ratio between consecutive terms.

c Write a simplified expression for the nth term of the sequence, T_n.

d Find the 10th term of the sequence.

36 The plotted points represent terms in a geometric sequence:



a Complete the table of values for the given points.

n	1	2	3
T_n			

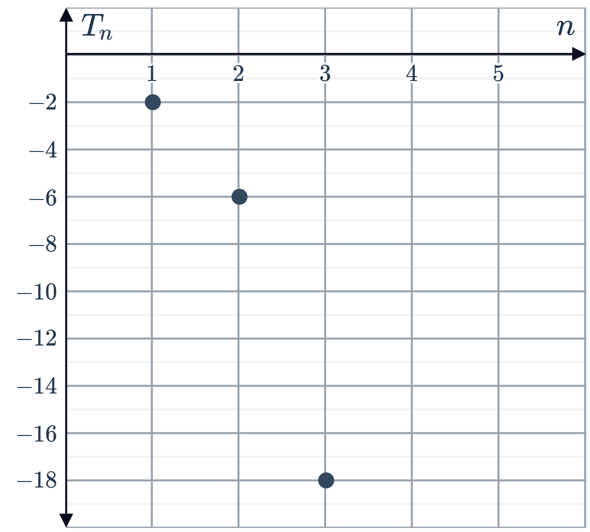
b Find r, the common ratio between consecutive terms.

c Find the 4th term of the sequence.

d Write a simplified expression for the nth term of the sequence, T_n.

e Find the 10th term of the sequence.

37 The plotted points represent terms in a geo[redacted]c sequence:



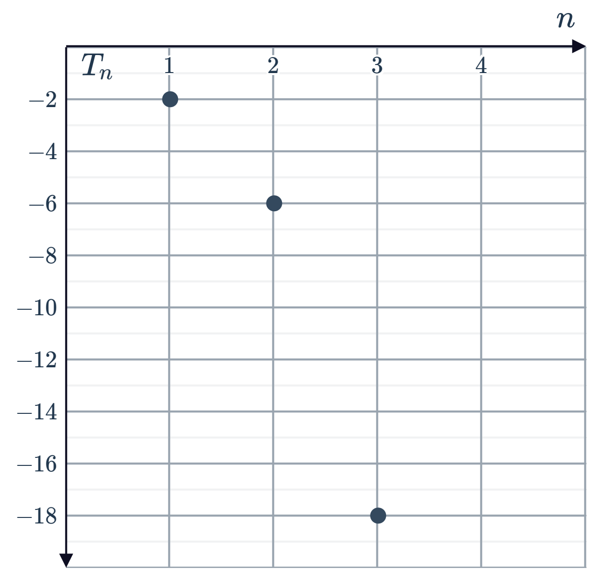
a Complete the table of values for the given points:

n	1	2	3
T_n			

b Find r, the common ratio between consecutive terms. Assume all values in the series are negative.

c Write a simplified expression for the nth term of the sequence, T_n.

38 The plotted points represent terms in a geometric sequence:

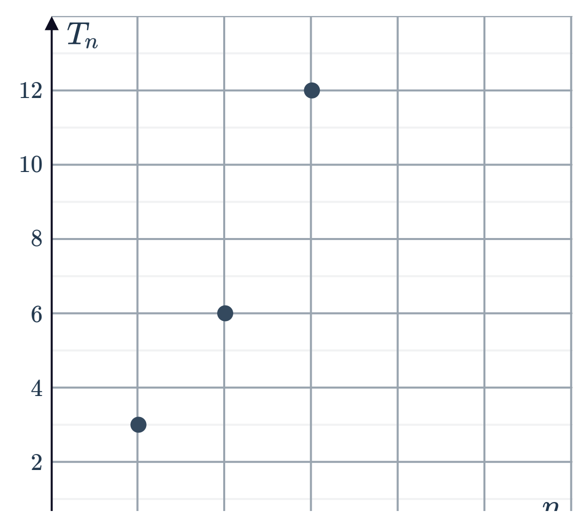


a Find the first term in the sequence.

b Find r, the common ratio between consecutive terms.

c Write a simplified expression for the nth term of the sequence, T_n.

39 The plotted points represent terms in a geometric sequence:

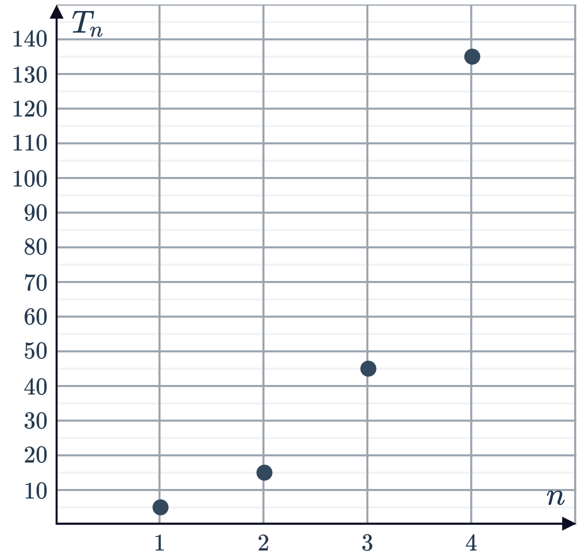


a Find r, the common ratio between consecutive terms.

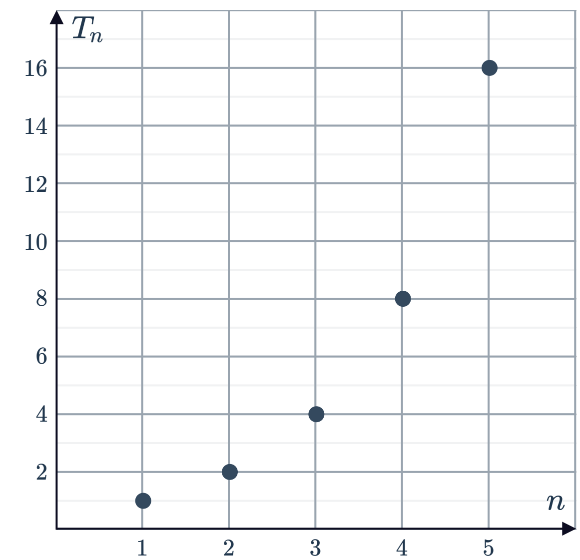
b Write a simplified expression for the nth term of the sequence, T_n.

c The points are reflected about the horizontal axis to form three new points. If these new points represent consecutive terms of a geometric sequence, write the equation for T_k, the kth term in this new sequence.

40 The plotted points represent terms in a geometric sequence:
Write a recursive rule for T_n in terms of T_{n-1} and an initial condition for T_1.



41 Consider the sequence plot drawn below:



a State the terms of the first five points of the sequence.

b Is the sequence arithmetic or geometric? Explain your answer.

c Write a recursive rule for T_{n+1} in terms of T_n and an initial condition for T_1.

Applications

42 The average daily growth of a seedling is 6% per day. A seedling measuring 8 cm in height is planted.

a Find the height of the seedling at the end of day 1.

b Find the height of the seedling, to two decimal places, 4 days after it is planted.

c Write a recursive rule, H_n, defining the height of the seedling n days after it is planted.

43 The average rate of depreciation of the value of a Ferrari is 14% per year. A new Ferrari is bought for $\$90\,000$.

- **a** Calculate the value of the car after 1 year.
- **b** Calculate the value of the car after 3 years.
- **c** Write a recursive rule, V_n, defining the value of the car after n years.

44 The average annual rate of inflation in Kazakhstan is 2.6%. Bread cost $\$3.65$ in 2015.

- **a** What would the bread cost in 2016?
- **b** At this rate, what would bread cost in 2018?
- **c** Write a recursive rule, V_n, defining the cost of bread n years after 2015.

45 Suppose you save $\$1$ the first day of a month, $\$2$ the second day, $\$4$ the third day, $\$8$ the fourth day, and so on. That is, each day you save twice as much as you did the day before.

- **a** How much will you put aside for savings on the 17th day of the month?
- **b** How much will you put aside for savings on the 29th day of the month?
- **c** How much will your total savings be for the first 13 days?
- **d** How much will your total savings be for the first 29 days?

46 This year, 600 people are expected to enter the workforce as registered nurses. This number is expected to increase by 4% next year, and increase by the same percentage every year after that.
Calculate the following, rounding your answers to the nearest whole number:

- **a** The number of nurses expected to enter the workforce between six and seven years from now.
- **b** The number of nurses expected to enter the workforce over the next six years.

47 A sample of 2600 bacteria was taken to see how rapidly the bacteria would spread. After 1 day, the number of bacteria was found to be 2912.

- **a** By what percentage had the number of bacteria increased over a period of one day?
- **b** If the bacteria continue to multiply at this rate each day, what will the number of bacteria grow to eighteen days after the sample was taken? Round your answer to the nearest whole number.

48 A rectangular poster originally measures 81 [redacted] width and $256\ \text{in}$ in length. To edit the poster once, the length of the rectangle is decreased by $\frac{1}{4}$ and the width is increased by $\frac{1}{3}$.

a If the poster is edited once, find the ratio of the original area of the rectangle to the new area.

b If the edit is repeated 3 times, calculate the new area of the poster to the nearest square inches.

c Find the number of times, n, that the process must be repeated to produce a square poster.

49 A conveyor belt is being used to remove materials from a quarry. Every thirty minutes, the conveyor belt empties out $\frac{1}{5}$ of whatever material remains in the quarry.
The quarry initially holds $14\,000\ \text{yd}^3$ of materials.

a How much material has been pumped out after 90 minutes?

b How much material is left in the dam after 90 minutes?

50 To test the effectiveness of a new antibiotic, first a certain bacteria is introduced to a body and the number of bacteria is monitored. Initially, there are 19 bacteria in the body, and after four hours, the number is found to double.

a If the bacterial population continues to double every four hours, how many bacteria will there be in the body after 24 hours?

b The antibiotic is applied after 24 hours, and is found to kill one third of the germs every two hours. How many bacteria will there be left in the body 24 hours after applying the antibiotic?
Assume the bacteria stops multiplying and round your answer to the nearest integer if necessary.

51 The annual output of a coal mine decreases by twenty percent each year. The output in the first year is $274\,000\ \text{m}^3$.

a Find the total amount of output in the first 8 years, to two decimal places if needed.

b Assuming there is always enough coal in the mine for the plant's operations, find the limit to this plant's total output production, to the nearest cubic meter.

52 The zoom function in a camera multiplies the dimensions of an image. In an image, the height of waterfall is $30\ \mathrm{mm}$. After the zoom function is applied once, the height of the waterfall in the image is $36\ \mathrm{mm}$. After a second application, its height is $43.2\ \mathrm{mm}$.

- **a** Each time the zoom function is applied, by what factor is the image enlarged?
- **b** If the zoom function is applied a third time, find the exact height of the waterfall in the image.

53 The first blow of a hammer drives a post a distance of $64\ \mathrm{in}$ into the ground. Each successive blow drives the post $\frac{3}{4}$ as far as the preceding blow. In order for the post to become stable, it needs to be driven $\frac{781}{4}\ \mathrm{in}$ into the ground.
If n is the number of hammer strikes needed for the pole to become stable, find n.